

CS144 Course Intro (Fall 2026)



Keith Winstein



Wireless Communications:
1964

*"A network to
survive nuclear attack."*

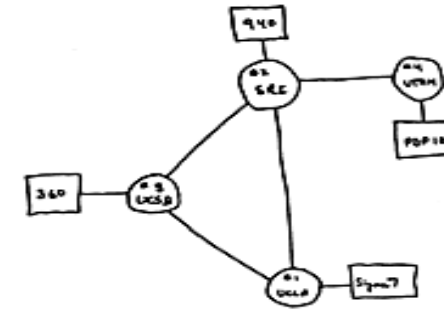


Paul
Baran

1st network
connects two
computers

US Government
starts "ARPANET"
project

Four nodes connected
(UCLA, SRI, UCSB, Utah)



1964

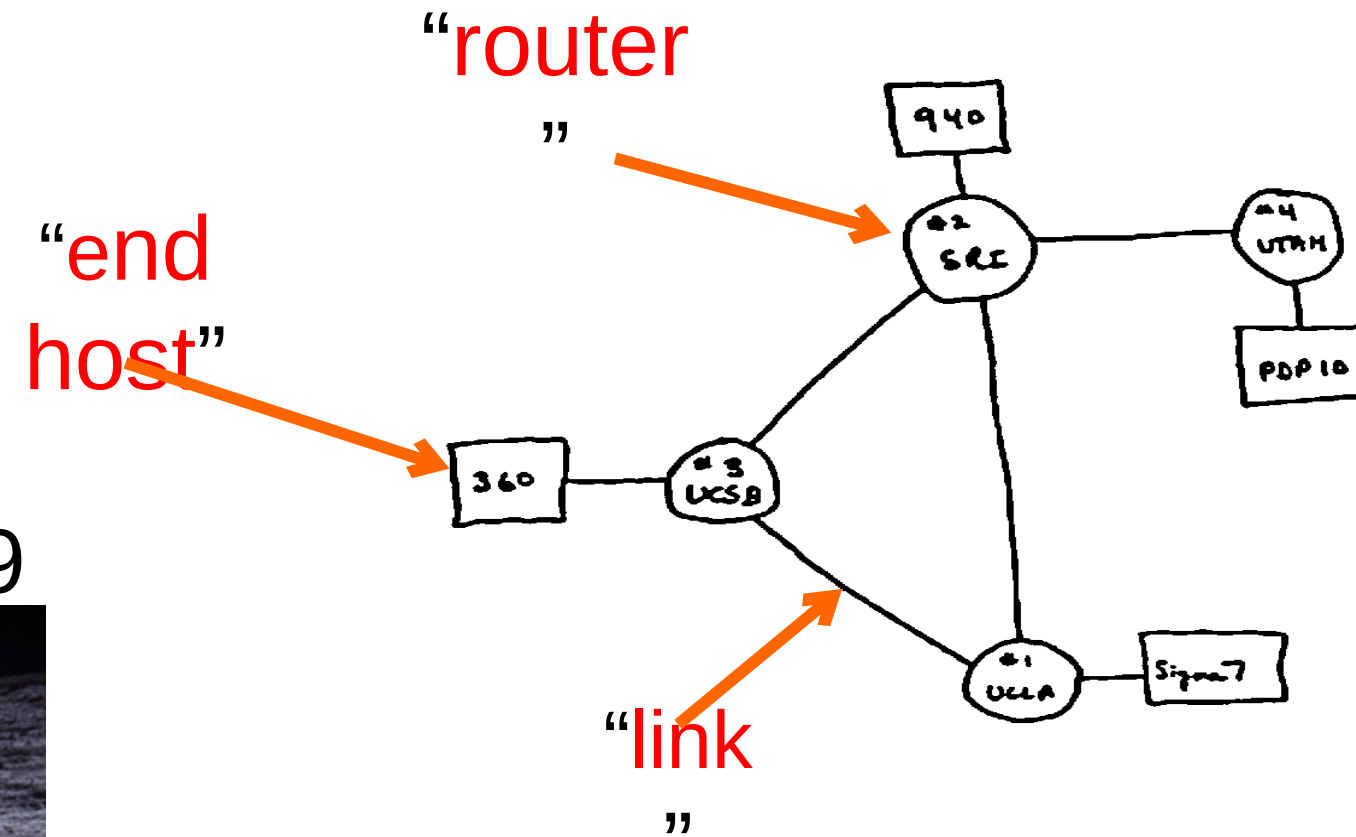
1965

1966

1968

1969

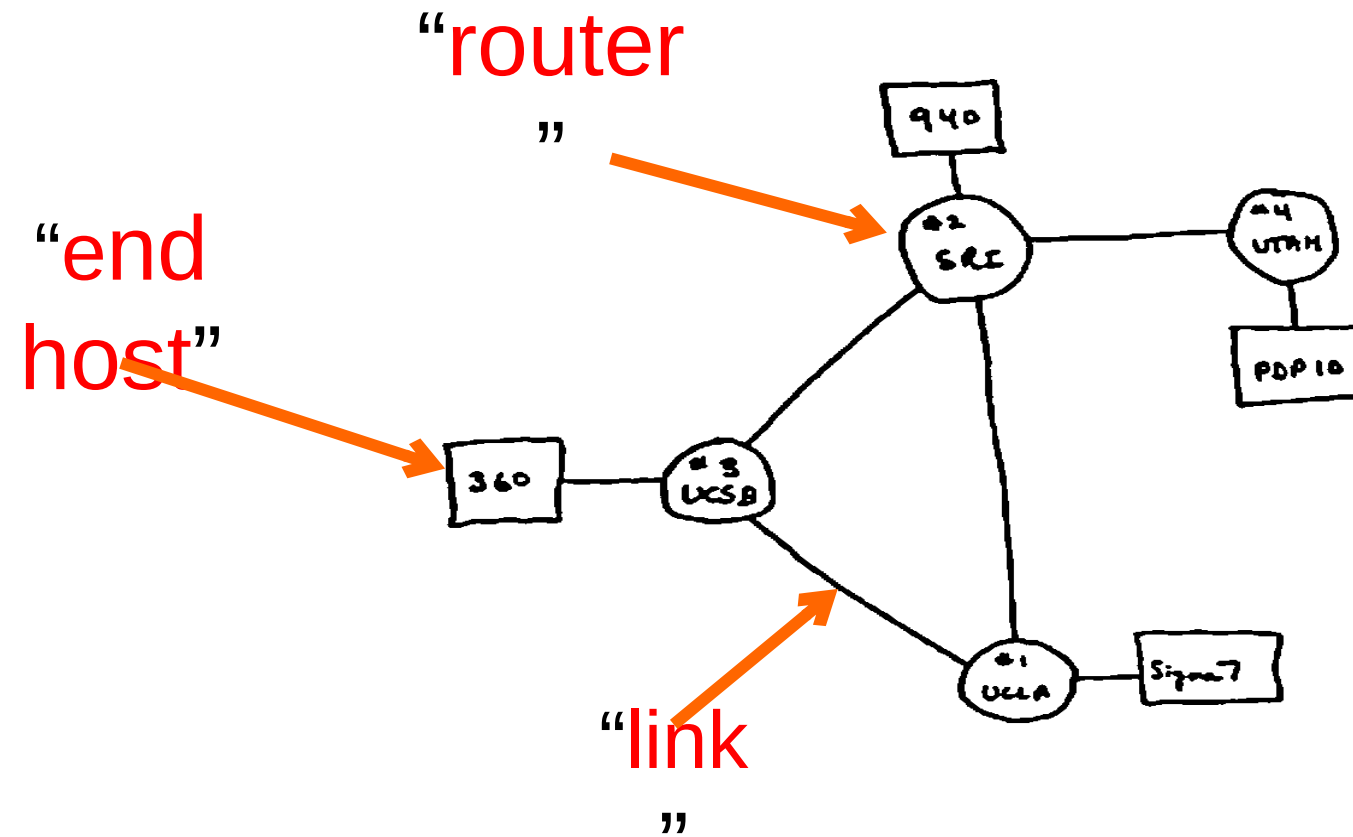
The Internet in 1969



Also in 1969

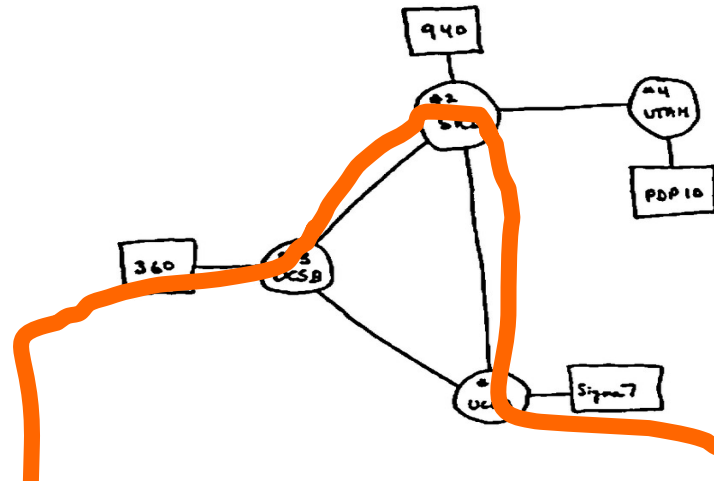


What did they use it for?



1. Sending files between scientists: *"Here is a big file of astronomy data!"*
2. Email: *"Where shall we have lunch today?"*
3. Remote login to another computer.

197
1

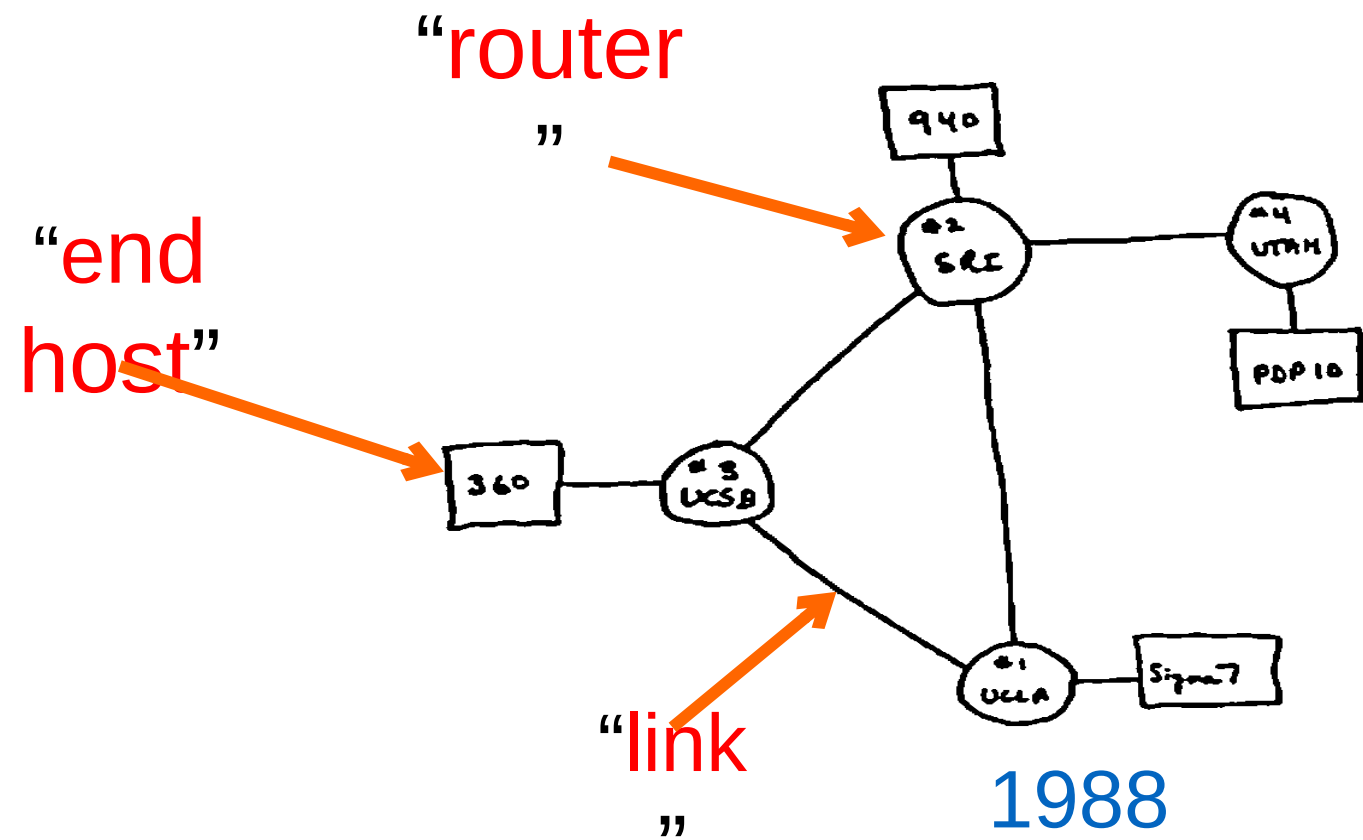


First email typed here

“QWERTYUIOP”

...and printed here





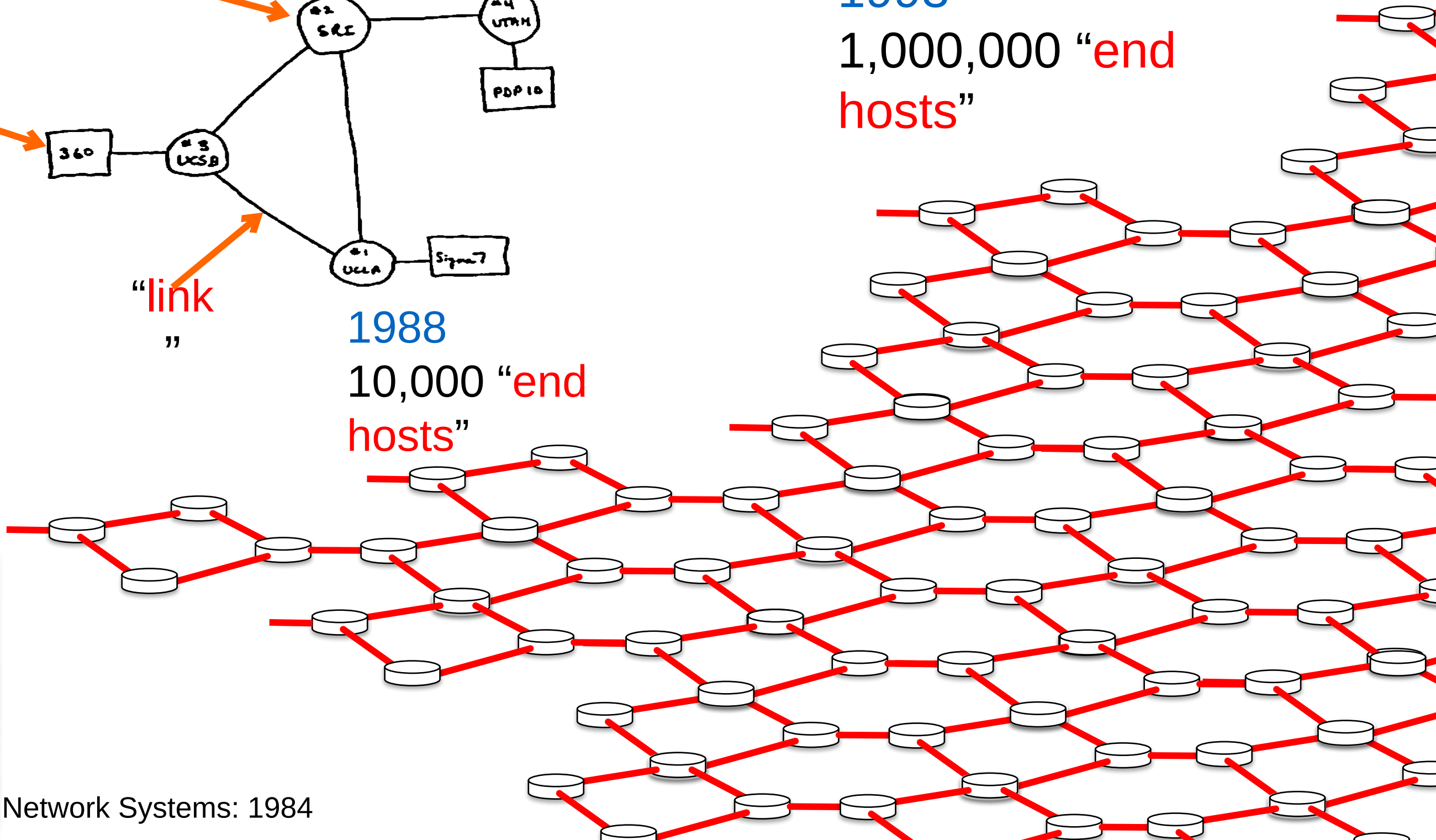
1969
4 “end
hosts”



Network Systems: 1984

1988
10,000 “end
hosts”

1993
1,000,000 “end
hosts”



Then in 1993 something
even BIGGER happened!!!

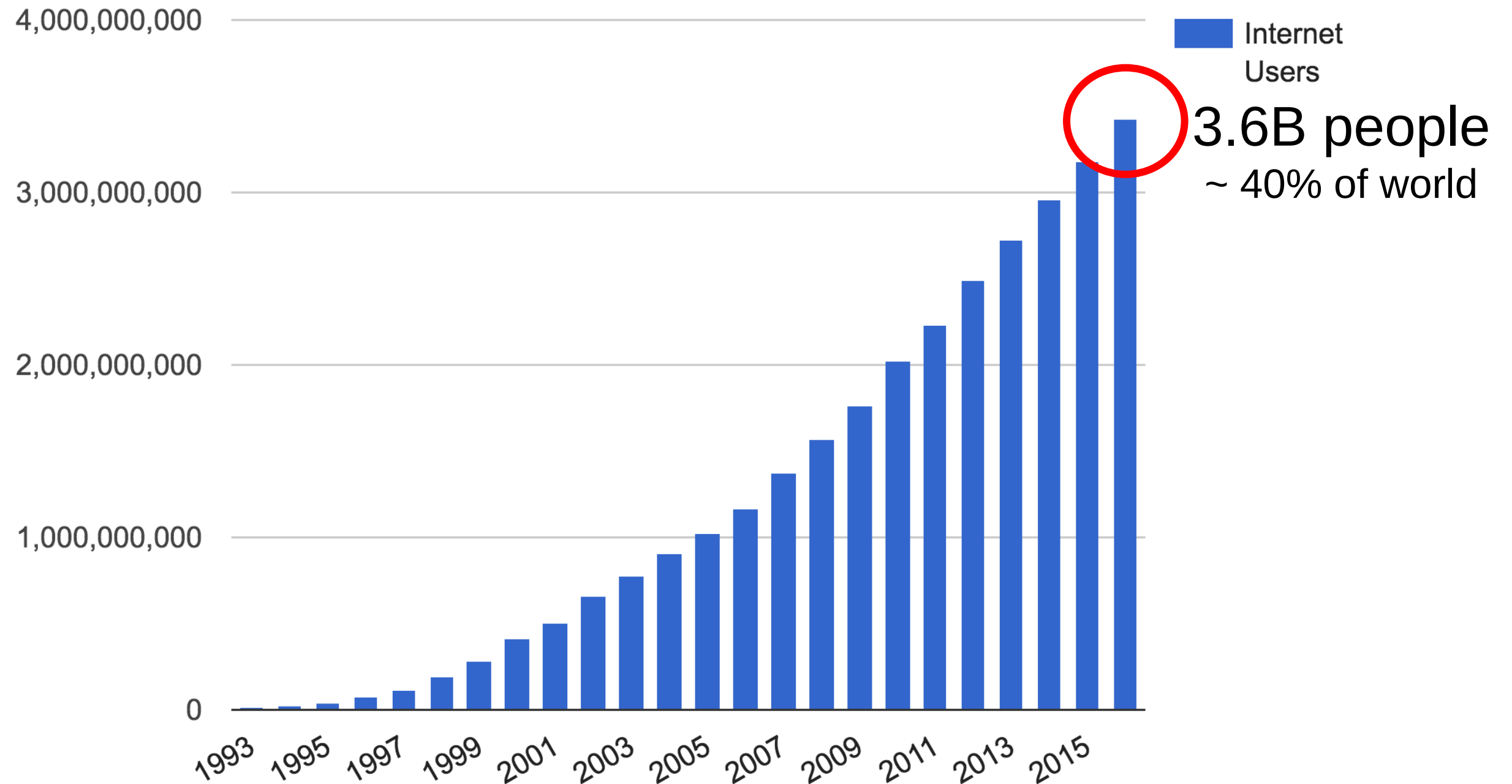
1993: The first web browser (Mosaic)



Marc Andreessen



The number of Internet users in the world



Source:

<http://www.internetlivestats.com/>



How does it all work?

Why was it designed this way?

CS144

Isn't it really difficult....?

Goals

1. To learn how the Internet works.
2. To learn why the Internet was designed this way.
3. To learn how to use the Internet.
4. To build some significant pieces of the Internet.
5. To learn the underlying principles and technologies of networking.

Class overview

CS144 is a fast-paced lab class.

Each week has TWO lectures, ONE lab session, and ONE tutorial.

The is divided into week-long units, devoted to a particular topic.

e.g. the 1st week is about Basic Principles, then (2) Transport, (3) Packet Switching, (4) Congestion Control, (5) the practical Internet, (6) Applications, (7) Routing, ...

There is ONE BIG LAB ASSIGNMENT (“you build most of the Internet”) with 8 “checkpoints,” #0 through #7.

Laptops

- We discourage laptop use in class, especially use that will distract others behind you.
- Except for specific in-class exercises (we will ask you to bring your laptop).



How we calculate your grade

(weights on website)

Q: “What does it take to earn an A?”

Common questions

Q: Can I take CS144 without the CS111 prereq?

A: We really don't encourage it! Ultimately prereqs are enforced by you, and if you choose to waive this rule, please don't make yourself a burden on the teaching staff.

Q: Can I take a conflicting class?

A: We really don't encourage it! You will be more likely to learn the material for real if you're here in lecture. Ultimately this is enforced by you, and if you choose to waive this rule, please don't make yourself a burden on the teaching staff (don't ask for a special final administered for you).

Exam Policy

Exams are closed-book, closed-note, closed-laptop etc.

But you may bring 1 double-sided piece of 8.5" x 11" paper of your own design to the Midterm, and 2 double-sided pieces of paper to the Final.

COVID

This quarter may ~~suck~~ present lousy circumstances. :-(

My job is to help you learn. I really want that to happen.

We're going to have to react to curveballs and we will do our best.

If you have to quarantine, we will be there for you (online) to help you avoid falling behind.

Lab

- Programming is in modern C++
- CS111 and C++ comfort are prerequisites.
- You want to go to the lab sessions + have started already
- Attendance is required:
 - For 4/9 lab sessions, and
 - any lab session if you're behind on the checkpoints
- Deadline policy:
 - Lab checkpoints are due Sundays, 11:59 p.m. (except week 6)
 - If you turn checkpoint in on time, we will get you subjective feedback in your tutorial (before next checkpoint is due).
 - Late/extension deadline is Wednesday, 7 p.m. (just before lab session). We can't guarantee a grading timeline for these, but you can always ask for feedback at lab session.
 - Functionality/correctness: no-penalty regrades until end of class.

Lab grading

- New this year: the “subjective” portion of the lab grade will be autograded to free the CAs up to run tutorials
- The autograder will be a reproducible deterministic process (not an LLM service, but may include a local deterministic LLM).
- Please bear with us as we get this running reliably; disagreements/regrade requests are welcomed and will be happily heard in lab.
- Regrade request will examine your whole submission (including writeup).



Win Certificates!!!

1. Certificate for BEST checkpoint submission
2. Certificate for BEST help (credited in lab writeups)
3. Certificate for BEST help (on Ed)

Workload



- This is a 4-unit workload, which means a workload of about 12hrs/week
- Our estimate based on previous years
 1. Lab+tutorial time: 3.7hrs/week
 2. Labs: Avg 6hrs/week
 - *wide variance between checkpoints and between students*
 - *read the whole checkpoint BEFORE coding*
 3. Average overall ~12hrs/week

Contact

For anything non-private: **Ed**

If private: Private Ed posting

(If it's a mega-emergency, Keith's cell phone number is on his Web page.)

The Honor Code

- We take it seriously and we expect you to take it seriously too.
- AI services have obviously made cheating more common. Generally students don't exactly set out to cheat: They had other priorities. They had a sick pattern. They asked friends for "help" and got tempted by their screens. Or they had other deadlines and found Claude Code (trained on our solutions) too tempting.
- Please, let's have a zero-violation year. The Honor Code was an amazing commitment by the students to govern and deter cheating themselves because you **don't** want the faculty to police you. Please show that student self-government and mutual deterrence works so we can keep faculty out of it.

The Honor Code

Permitted Collaboration: The following items are encouraged and allowed at all times for all students in this class:

- Discussion of material covered during lecture, problem sessions, or in handouts
- Discussion of the requirements of an assignment
- Discussion of the use of tools or development environments
- Discussion of general approaches to solving problems
- Helping others at the lab session (**without** looking at code!)
- Discussion of general techniques of coding or debugging

The Honor Code

Unpermitted Collaboration: All submissions must represent **original, independent work**. Some examples of activities that do not represent original work include:

- Copying solutions from others or knowingly allowing others to copy your solution.
- Use of solutions posted to websites.
- Looking at another student's code (unless handwritten, not CS144-specific, **and** credited).
- Placing your source code in a public repository where others can copy it.
- Debugging code for someone else.
- Collaborating on or discussing the online graded quizzes before you have completed them.
- Using output from a code-synthesis tool (e.g. ChatGPT/Claude/Gemini/GLM/etc.) trained on past students' solutions.
- *Honor code exercise: can you stump the teaching staff?*

What to do next

- Look around and get familiar with <https://cs144.stanford.edu>
- Start setting up your VM for Checkpoint 0! Don't wait until tonight to start downloading (the Wi-Fi will be very congested if you do).
- See you tonight!

TCP/IP Header Formats in Lego

